MILITARY SPECIFICATION

COLLIMATOR, INFINITY AIMING REFERENCE: M1

This amendment forms a part of Military Specification MIL-C-46372A(MU), dated 27 February 1964.

Paragraph 2.1, SPECIFICATIONS: Delete "MIL-I-45208".

Paragraph 2.1, DRAWINGS: Delete "IEL8644275" and "LSQAP8644275".

* Paragraph 3.6.1: Delete and substitute:

"3.6.1 Collimator unit (telescope).- The collimator unit shall be capable of holding an internal pressure of dry nitrogen of 5 psi for a period of 2 hours ±5 minutes without loss of pressure in excess of 0.2 psi for the 2 hour period. The internal pressure of the collimator shall then be reduced to zero psig. The collimator shall then be flushed with dry nitrogen until the interior atmosphere has a dew point at least as low as minus 25°F. The collimator shall then be sealed with an internal pressure of between 0.25 and 0.50 psig of dry nitrogen."

Paragraph 3.9.3.1: Add the following:

"With the azimuth fine adjustment knob at its extreme travel position (both in and out), the adjustment assembly shall withstand a torque load of 25 inch pounds plus 20 percent, without stripping the threads within the adjustment assembly when checked at standard ambient temperature, and when exposed to, and checked at the extreme temperatures of 3.4.2."

Paragraph 4.1.1: Delete and substitute:

"4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109."

Add the following paragraph:

"4.1.2 First article (initial production) inspection.- When required by the contract, an initial production sample shall be submitted to a Government approved facility in accordance with the contract schedule for test and evaluation. The sample shall consist of three collimators plus three of every component and subassembly covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the

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same manner, using the same materials, equipment, processes, and procedures, as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. After inspection by the contractor and provisional acceptance at source by the Government, the sample shall be inspected at a Government facility or such other facility as specified in the contract, for all the requirements of the drawings and specifications."

Paragraph 4.2.2: Delete the 1st sentence and substitute:

"All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply." Delete the 4th and 5th sentences and substitute: "The tabulated classification of defects in Table II and 4.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot."

Paragraph 4.4: Delete and substitute:

"4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607."

Paragraph 4.4.1: Delete and substitute:

"4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607."

Paragraphs 4.4.2, 4.4.2.1 and 4.4.2.2: Delete and substitute:

"4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements

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of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment."

* Paragraph 4.5.3.1: Delete and substitute:

"4.5.3.1 Collimator unit (telescope).- Internal pressurizing of the collimator to test for sealing shall be performed using nitrogen gas having a dew point at least as low as -25°F, and commercial type pressure testing equipment capable of pressurizing the collimator to the pressure specified in paragraph 3.6.1. The test equipment shall be equipped with pressure shut-off valve, filter, and a calibrated pressure gage to read required pressure value specified in 3.6.1 to an accuracy of ± .02 psi. Maintain the pressure within collimator for the period specified. Any drop in pressure as indicated on the pressure gage of the test equipment shall not exceed requirement specified in 3.6.1. The pressure shall then be reduced to zero psig (normal atmospheric pressure). The collimator shall then be flushed with nitrogen to the requirements of 3.6.1. The interior atmosphere shall be periodically sampled through a dew point meter until the atmosphere shows a dew point at least as low as -25°F. When this atmosphere is attained within the collimator, the collimator shall be sealed with an internal pressure of nitrogen at a pressure of between 0.25 and 0.50 psig at room temperature."

Paragraph 4.5.6.5: Add the following:

"The torque withstanding load test of the azimuth fine adjustment at each travel stop position (in and out), shall be measured with standard measuring equipment attached to the knob. At each temperature specified in 3.9.3.1 rotate the azimuth fine adjustment to the travel stop position, then apply the specified 25 inch pound torque load to the knob. During this torque application there shall be no evidence of knob rotation. Upon return to standard ambient temperature, and subsequent to these torque applications the azimuth fine adjustment knob shall be manually rotated in and out to each stop. During these rotations there shall be no tactile evidence of damaged threads. The azimuth fine adjustment shall conform to the requirements of 3.9.3.1."

*The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to

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evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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